Light

GCSE Computer Science – Data Representation: Numbers (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset ← Back to Data Representation Numbers

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which digits can appear in a binary number?

- **A.** 0 and 1 only
- **B.** 0 to 7
- **C.** 0 to 9
- **D.** 0 to 9 and A to F

Q2 (1 mark)

Why do programmers often use hexadecimal instead of binary?

- **A.** Hexadecimal numbers are processed faster by the CPU

- ○ **B.** Hexadecimal is shorter and easier to read while converting directly to binary
- ○ **C.** Hexadecimal can represent negative numbers but binary cannot
- ○ **D.** Hexadecimal is the native language of all computer processors

Q3 (1 mark)

What is the denary value of the binary number 10011?

- ○ **A.** 17
- ○ **B.** 18
- ○ **C.** 19
- ○ **D.** 20

Q4 (1 mark)

What is the binary representation of the denary number 13?

- ○ **A.** 1011
- ○ **B.** 1100
- ○ **C.** 1110
- ○ **D.** 1101

Q5 (1 mark)

In hexadecimal, what denary value does the letter D represent?

- ○ **A.** 13
- ○ **B.** 12
- ○ **C.** 14
- ○ **D.** 4

Q6 (1 mark)

What is the largest value that can be represented by a single hexadecimal digit?

- ○ **A.** 9
- ○ **B.** 15
- ○ **C.** 16
- ○ **D.** F

Q7 (1 mark)

What is a nibble in computing?

- ○ **A.** 8 bits
- ○ **B.** 4 bits
- ○ **C.** 16 bits
- ○ **D.** 2 bits

Q8 (1 mark)

What is the maximum denary value that can be stored in 6 bits?

- ○ **A.** 32
- ○ **B.** 64
- ○ **C.** 31
- ○ **D.** 63

Q9 (1 mark)

In an 8-bit binary number, which bit is the least significant bit (LSB)?

- ○ **A.** The rightmost bit
- ○ **B.** The leftmost bit
- ○ **C.** The bit with value 128
- ○ **D.** The middle bit

Q10 (1 mark)

In binary addition, what is the result of 1 + 1 + 1 (three ones)?

- ○ **A.** 2
- ○ **B.** 10
- ○ **C.** 11

- ○ **D.** 100

Q11 (1 mark)

What happens to the value of a binary number when it is shifted left by one place?

- ○ **A.** The value is halved
- ○ **B.** The value increases by 1
- ○ **C.** The value is doubled
- ○ **D.** The value stays the same

Q12 (1 mark)

What is an overflow error in binary arithmetic?

- ○ **A.** When a calculation produces a negative result
- ○ **B.** When dividing by zero
- ○ **C.** When converting between number bases incorrectly
- ○ **D.** When a calculation result is too large to fit in the available number of bits

Q13 (2 marks)

Convert the denary number 173 to an 8-bit binary number.

- A. 10101011
- B. 10101101
- C. 10101111
- D. 10110101

Q14 (2 marks)

Convert the binary number 11001110 to denary.

- A. 198
- B. 202
- C. 206
- D. 210

Q15 (2 marks)

Convert the hexadecimal number 5A to denary.

- A. 80
- B. 85
- C. 95
- D. 90

Q16 (2 marks)

Convert the denary number 158 to hexadecimal.

- ○ **A.** 9E
- ○ **B.** 9D
- ○ **C.** A8
- ○ **D.** AE

Q17 (2 marks)

Convert the binary number 11010110 to hexadecimal.

- ○ **A.** C6
- ○ **B.** DA
- ○ **C.** D6
- ○ **D.** E6

Q18 (2 marks)

Convert the hexadecimal number B3 to binary.

- ○ **A.** 10110011
- ○ **B.** 10110001
- ○ **C.** 10110010

- ○ **D.** 10110100

Q19 (2 marks)

What is the result of the binary addition 01011010 + 00110101?

- ○ **A.** 10001101
- ○ **B.** 10011111
- ○ **C.** 10001111
- ○ **D.** 10010001

Q20 (2 marks)

Which of these 8-bit binary additions would cause an overflow error?

- ○ **A.** 01100100 + 01100100
- ○ **B.** 10000000 + 01111111
- ○ **C.** 01111111 + 00000001
- ○ **D.** 11000000 + 01000000

Q21 (2 marks)

The binary number 00001011 is shifted left by 2 places. What is the result and what multiplication does this represent?

- ○ **A.** 00101100, multiplied by 4

- B. 00101100, multiplied by 2
- C. 01011000, multiplied by 2
- D. 01011000, multiplied by 8

Q22 (2 marks)

What is the result when binary 01100100 is shifted right by 2 places?

- A. 00011000
- B. 00011001
- C. 10011001
- D. 00110010

Q23 (2 marks)

How many bits are in 2 kilobytes (KB)?

- A. 2,000 bits
- B. 2,048 bits
- C. 16,384 bits
- D. 16,000 bits

Q24 (3 marks)

A computer performs the binary addition 11011011 + 01010101.

Analyse the calculation, showing the result and explaining whether overflow occurs in 8 bits.

- ○ **A.** Result is 100110000; overflow occurs because 9 bits are needed, losing the leftmost bit to give 00110000
- ○ **B.** Result is 00110000; no overflow because the result fits in 8 bits
- ○ **C.** Result is 11111110; overflow occurs because carries propagate
- ○ **D.** Result is 100110000; no overflow because the sum is less than 256

Q25 (3 marks)

The binary number 10110111 is shifted right by 3 places.
Analyse what happens to the value, calculating the expected mathematical result and the actual stored result.

- ○ **A.** Expected 22.875, stored 23; rounding up occurs automatically
- ○ **B.** Expected 22.875, stored 22; three bits (111) lost causing truncation
- ○ **C.** Expected 23, stored 23; no data loss occurs
- ○ **D.** Expected 22, stored 22; the result is exact with no remainder

Q26 (3 marks)

A web designer uses the hexadecimal colour code #00FF00.
Analyse what colour this represents and explain the structure of RGB colour codes.

- ○ **A.** Pure red; FF in the red channel with 00 in green and blue
- ○ **B.** Yellow; FF in both red and green channels

○ **C.** Pure green; FF in the green channel (255) with 00 in red and blue (0 each)

○ **D.** Cyan; FF in both green and blue channels

Q27 (3 marks)

A temperature sensor records values from -40°C to +50°C. A student suggests using unsigned 8-bit binary.
Analyse why this is problematic and determine the minimum bits needed for signed representation.

○ **A.** Unsigned works fine; the sensor should add 40 to all readings

○ **B.** Unsigned is correct; negative temperatures are stored as values above 128

○ **C.** Unsigned cannot store values above 50; 16-bit is required

○ **D.** Unsigned cannot store negative values; 8-bit signed (-128 to +127) would work

Q28 (3 marks)

Convert the denary number 227 to binary and hexadecimal.
Verify your answer by converting the binary back to denary.

○ **A.** Binary: 11100011, Hex: E3; Verification: 128+64+32+2+1 = 227

○ **B.** Binary: 11100011, Hex: E3; Verification: 128+64+32+4+1 = 229

○ **C.** Binary: 11100101, Hex: E5; Verification: 128+64+32+4+1 = 229

○ **D.** Binary: 11100001, Hex: E1; Verification: 128+64+32+1 = 225

Q29 (3 marks)

A programmer needs to multiply a number by 10 using only binary shifts and addition.
Analyse how this can be achieved and explain the method.

- ○ **A.** Shift left 10 times to multiply by 10 directly
- ○ **B.** Shift left 3 times (x8) plus shift left 1 time (x2), then add results: 8x + 2x = 10x
- ○ **C.** Shift left 3 times (x8) plus the original twice: 8x + x + x = 10x
- ○ **D.** It is impossible to multiply by 10 using only shifts and addition

Q30 (3 marks)

A music file is 4 MB (megabytes). A streaming service delivers audio at 320 kbps (kilobits per second).
Analyse how long it takes to stream the file, noting the conversion between bits and bytes.

- ○ **A.** 12.5 seconds; divide 4 MB directly by 320 kbps
- ○ **B.** 1,000 seconds; 4 MB / 4 kbps after adjusting for bit/byte difference
- ○ **C.** 100 seconds; convert MB to bits (32,000,000 bits), divide by 320,000 bps
- ○ **D.** 800 seconds; multiply MB by 8 then divide by kbps

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.